DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING
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Experiment No.: 1

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Branch: AIT-CSE **(AI ML)** **Section/Group:** B
Semester: 5TH **Date of Performance:**
Subject Name: Competitive Coding – II **Subject Code:** 24CSP-305

Experiment No: 1.1

Aim: Contains Duplicate II

Objective: To find whether duplicate elements exist within a distance of k indices.

Problem Statement:
Given an array nums and an integer k, return true if the same number appears at two different indices whose difference is at most k.

LeetCode Link: https://leetcode.com/problems/contains-duplicate-ii/?utm_source=chatgpt.com

Procedure/Algorithm/Pseudocode:

1. Create a HashMap.
2. Traverse the array.
3. Check if the current number already exists.
4. If the index difference is <= k, return true.
5. Otherwise, store the current index.
6. Return false.

Code:

```
class Solution {

public:

   bool containsNearbyDuplicate(vector<int>& nums, int k) {

      unordered_map<int, int> lastSeen;

      for (int i = 0; i < nums.size(); i++) {

         // Check if number was seen before

         if (lastSeen.find(nums[i]) != lastSeen.end()) {

            if (i - lastSeen[nums[i]] <= k) {

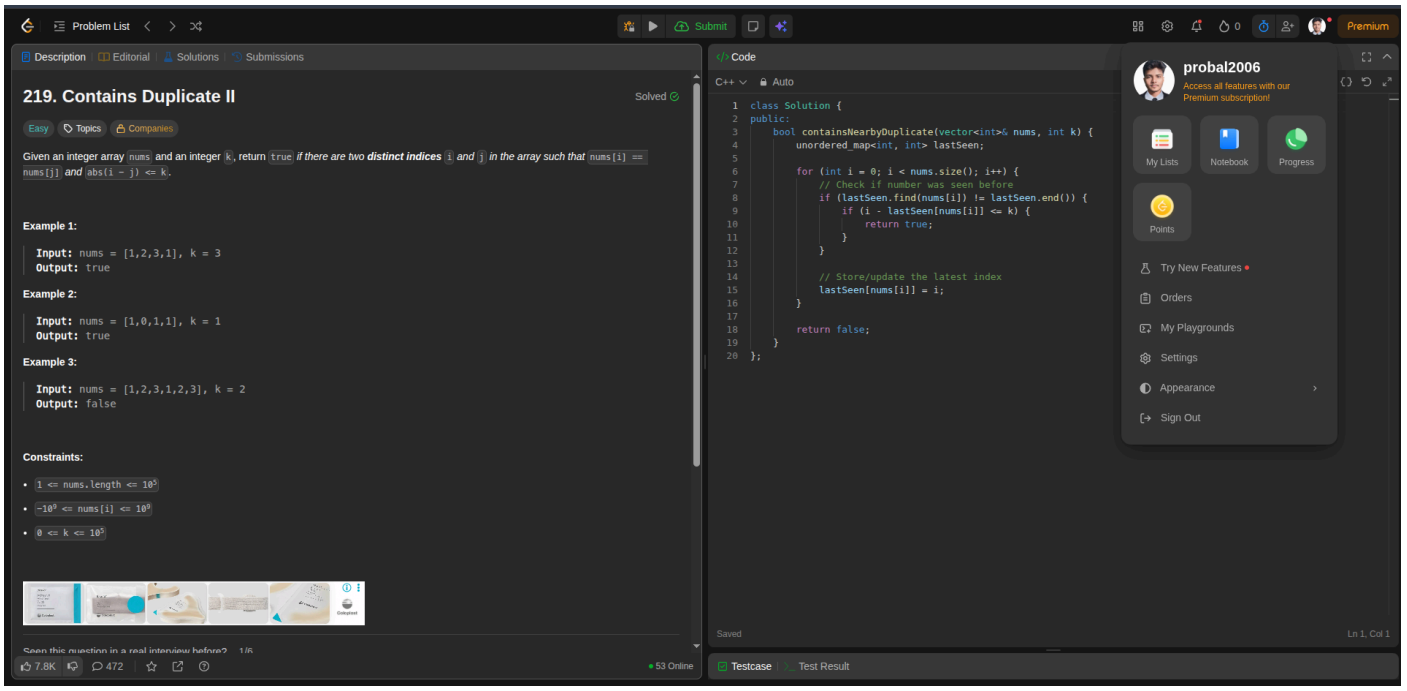
               return true;

            }

         }
```

```
            // Store/update the latest index
            lastSeen[nums[i]] = i;
        }
        return false;
    }
};
```

Output:



Observations/Outcome:

The program successfully detects duplicate values within the given index distance.

Time Complexity: O(n)

Space Complexity: O(n)

Learning outcomes:

1. Learned to use HashMap.
2. Learned duplicate detection.
3. Learned index distance calculation.
4. Learned optimized searching.
5. Learned time and space complexity analysis.

Experiment No: 1.2

Aim: Product of Array Except Self

Objective: To calculate the product of all elements of an array except the element at the current index.

Problem Statement:

Given an integer array nums, return an array answer where:

answer[i] = product of all elements except nums[i]

The solution should not use division.

LeetCode Link: https://leetcode.com/problems/product-of-array-except-self/?utm_source=chatgpt.com

Procedure/Algorithm/Pseudocode:

1. Create an output array.
2. Calculate the prefix product for every element.
3. Calculate the suffix product from right to left.
4. Multiply prefix and suffix products.
5. Return the output array.

Code:

```
class Solution {

public:

   vector<int> productExceptSelf(vector<int>& nums) {

      int n = nums.size();

      vector<int> answer(n, 1);

      // Prefix product

      int prefix = 1;

      for (int i = 0; i < n; i++) {

         answer[i] = prefix;

         prefix *= nums[i];

      }

      // Suffix product

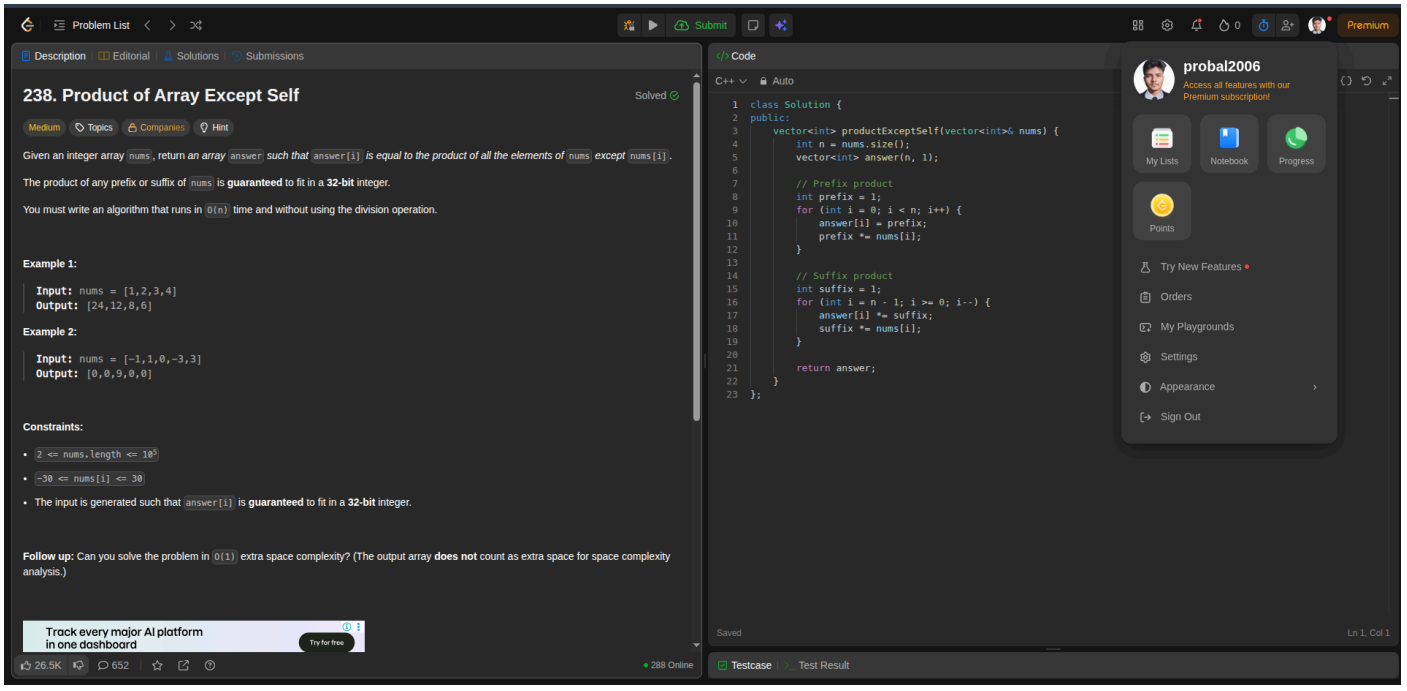
      int suffix = 1;

      for (int i = n - 1; i >= 0; i--) {

         answer[i] *= suffix;
```

```
            suffix *= nums[i];
        }
        return answer;
    }
};
```

Output:



Observations/Outcome: The program successfully calculates the product of all elements except the current element without using division.

Time Complexity: O(n)

Space Complexity: O(1)

Learning outcomes :

1. Learned the prefix product technique.
2. Learned the suffix product technique.
3. Learned to solve array problems without division.
4. Learned how to optimize space complexity.
5. Learned to analyze time and space complexity.